

Python Programming Assignments - 8

(1) Explain the steps to create and use a user-defined module.

→ A user-defined module is a python file (demo.py) that contains functions, variable or classes that you create & reuse in other python programs.

(a) create python file → demo.py have multiple function

(b) create another file → test.py in this will import entire module.

import demo → access the module contents the dot(.) operator

(2) why is indentation mandatory in python while it optional in languages like (C, C++, Java)?

→ Indentation defines a structure of programme and tells the interpreter ~~which~~ which statements belongs together.

(3) what is IndentationError? when does python raise this error

→ ~~python~~ Indentation Error occurs when spaces or tabs at the beginning of a line.

IndentationError Occurs:

```
def Ind_Error  
print("Hello")
```

(4) What are the different ways to import a module? Which one is recommended and why?

→ Different ways to import a module.

(a) Import the entire module

`import math`

(b) Import specific function & Variable

`from math import sqrt, pi`

(c) Import all

`from math import *`

(d) Import a module with a alias

`import math as m`

Recommend method is "import math" because it improves readability, avoid naming conflicts.

(5) How does python internally identify the start and end of a code block using indentation?

→ Python identifies the start and end of a code block by comparing the indentation level (number of spaces or tabs) of each line.

<6> why is from module import * discouraged in Professional code?

→ In this python copies all public function & variables from the module into your current namespaces. There is many chances to override the imported module functions.

<7> How does modular programming improve code reusability testing and maintenance?

→ modular programming breaking a large application down into smaller, independant chunk called modules - improves software quality accross three main pillars reusability, testing, and ~~maintenace~~ maintenance.

<8> what happens internally when python executes an import module_name statement?

→ when execute ^{import} module_name python combines two seperate operations it searches for the module and then binds the result to local variable name.

<9> what happens if a module contains print statements outside any function?

→ print statements will execute exactly once the very first time the module is imported in your program.

(10) Explain why python modules are considered the

Foundation of scalable software design?

→ modules break down massive, unmanageable code bases into small, independent and specialized components.

As a software project grows from hundreds of lines to millions, modules prevent the architecture from collapsing under its own weight.